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New discrimination method combining hit quality index based spectral matching and voting☆

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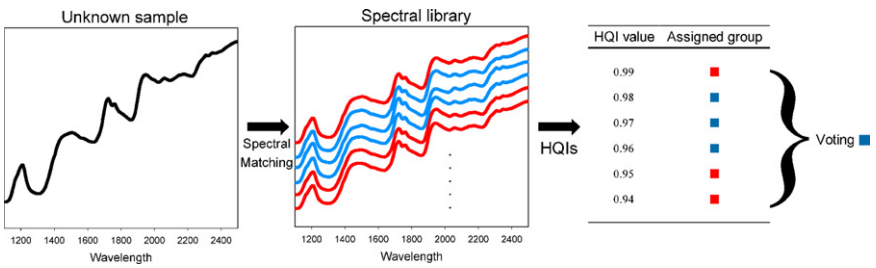
HIGHLIGHTS

- A discrimination method, called hit quality index (HQI)-voting, has been developed.
- It effectively utilizes HQI as a judgment factor for group determination.
- It is based on sample-to-sample spectral matching without modeling, so the model over-fitting is not an issue.
- It improved the discrimination for geographical origins of agricultural samples and similar petroleum products.

GRAPHICAL ABSTRACT



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ABSTRACT

A new discrimination method, called hit quality index (HQI)-voting, that uses the HQI for discriminant analysis has been developed. HQI indicates the degree of spectral matching between two spectra as known. In this method, a library sample yielding the highest HQI value for an unknown sample was initially searched and a group containing this sample was chosen as the group for the unknown sample. When overall spectral features of two groups are quite close to each other, many library samples with similar HQI values could be available for an unknown sample. In this situation, the simultaneous consideration of multiple votes (several library samples with close HQI values) for final decision would be more robust. In order to evaluate the discrimination performance of HQI-voting, three different near-infrared (NIR) spectroscopic datasets composed of two sample groups were used: (1) domestic and imported sesame samples, (2) domestic and imported Angelica gigas samples, and (3) diesel and light gas oil (LGO) samples. For the purpose of comparison, principal component analysis–linear discriminant analysis (PCA–LDA), partial least squares–discriminant analysis (PLS–DA) as well as *k*-nearest neighbor (*k*-NN) were also performed using the same datasets and the resulting accuracies were compared. The discrimination performances improved with the use of HQI-voting in comparison with those resulted from PCA–LDA and PLS–DA. The overall results support that HQI-voting is a comparable discrimination method to that of existing factor-based multivariate methods.

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1. Introduction

When vibrational spectroscopic methods, including near-infrared (NIR) spectroscopy, are used for the differentiation of samples into two groups, the use of multivariate discrimination methods is typical in order to extract relevant information

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from highly overlapping spectral features. Although diverse discrimination methods have been developed [1–6], three methods of principal component analysis–linear discriminant analysis (PCA–LDA) [2,3,7–9], partial least squares–discriminant analysis (PLS–DA) [2,3,10–12], and soft independent modeling of class analogy (SIMCA) [2,4,13–15] have been most frequently adopted for many practical field applications. In these methods, the determination of an optimal number of latent variables (factors) is important to avoid the over-fitting problem, always a highly concerning issue in factor-based multivariate analysis. In addition, a developed model does not explicitly explain how spectral features contribute in discriminant analysis, so commonly called a black-box model.

We proposed a discrimination method called hit quality index (HQI)-voting in this study, which effectively utilizes HQI as a classification measure. HQI is a numerical quantity that indicates the correlation between two spectra and has been widely used in the field of spectroscopy to indicate the degree of spectral matching in library searches [16–18]. A spectrum of an unknown sample is compared with all of the spectra of the known samples in a library and the best matching sample is found by examining the calculated HQI values. HQI provides a relative degree of spectral matching for an unknown sample with all of the samples in a library, but it does not provide statistical probability (or significance) of individual matching. A more detailed description of HQI can be found in relevant publications [16–18]. As described, HQI is a widely accepted method for spectral matching, so it could be reliably used as a spectral similarity measure for discriminant analysis.

Fig. 1 shows the graphical description showing the overall procedure of HQI-voting. Suppose a spectral library composed of n spectra with p-dimensions. For an unknown sample spectrum, the HQI values are calculated using Eq. (1) by individually matching it with every spectrum in a library and it results in n HQI values for an unknown sample.

$$\mathrm{HQI} = \frac{\left[\sum_{i=1}^{p}(A_i-\bar{A})(L_i-\bar{L})\right]^2}{\left[\sum_{i=1}^{p}(A_i-\bar{A})^2\right]\left[\sum_{i=1}^{p}(L_i-\bar{L})^2\right]} \quad (1)$$

In the equation, A and L are the vectors of the spectral responses of an unknown spectrum and library spectrum, respectively. Here, the sample with highest HQI value is the closest spectral match with an unknown sample. Then, as shown in Fig. 1, the samples in a library are sorted by the HQI values in descending order and a group containing a sample of the highest HQI value is assigned as the group for an unknown sample, black square group in this case. This example shows that only one sample that provides the highest HQI value is taken into consideration when determining the group.

When the overall spectral features between the two groups are quite similar to each other, several samples with close HQI values for an unknown sample could be available in a library. When determining a class (group) for an unknown sample in this situation, it is recommendable to consider multiple samples yielding high HQI values rather than only one sample with the closest match. For example, suppose that a group decision is made by taking into consideration 6 samples with the top 6 HQI values as shown in Fig. 1. After sorting HQI values for these samples, the sum of HQI for each class in a given k, alternatively the number of votes ($k=6$ here), is obtained. Here, the sum of white square group is greater than that of black square group, so the unknown sample is assigned to the white group. As shown, the result of classification could vary depends on the number of votes employed for the decision. In this method, an optimal number of voters in HQI-voting was decided at the point yielding a minimal misclassification rate (discrimination error) using a cross-validation method as routinely performed in many other studies.

As described above, the proposed method is analogous to k-nearest neighbor (k-NN) that classifies unknown samples based on their similarity (or similarity measure) with samples in a training set [19,20]. In k-NN, the use of odd number of neighbors (voter) is typical for classification; while, HQI-voting could further accommodate even number of voters without tie vote since the sums of HQI values voted for each class are compared in group determination. In addition, since the magnitude of HQI values is directly reflected, it imposes more weight on the opinions from samples with greater HQI values in classification.

In order to evaluate the discrimination performance of HQI-voting, three different spectral datasets composed of two sample groups were selected: (1) NIR diffuse reflectance spectra of domestic and imported sesame samples, (2) NIR diffuse reflectance spectra of domestic and imported Angelica gigas samples, and (3) NIR spectra of diesel and light gas oil (LGO) samples. For the purpose of comparison, PCA–LDA, PLS–DA as well as k-NN were also performed using the same datasets and the resulting discrimination accuracies were compared with those acquired using HQI-voting. In

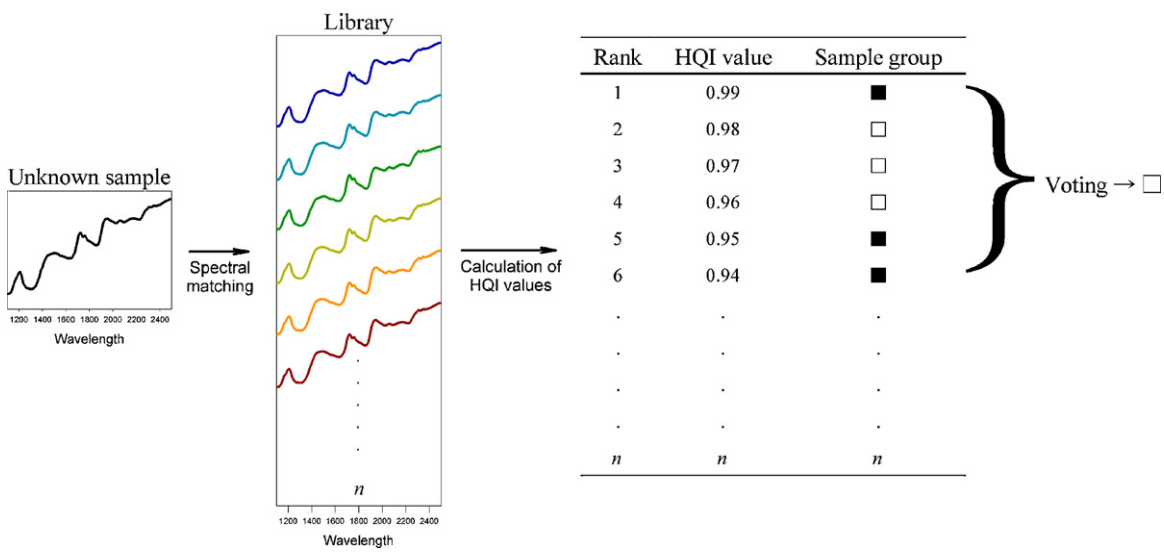

Fig. 1. Graphical presentation describing the overall procedure of HQI-voting for discrimination analysis.

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addition, the discrimination performance of HQI-voting was tested by artificially varying the number of calibration (library) samples in order to examine the dependence of their discrimination accuracy on the sample population.

2. Experimental

For analysis of the sesame samples, a diffuse reflectance NIR spectral dataset used in a previous publication [9] was used. It was composed of NIR spectra corresponding to 1122 sesame samples (432 imported and 690 domestic (Korean) samples) collected over 8 years in order to include diverse cultivation areas that ensure wider compositional variations. A diffuse reflectance NIR spectral dataset containing 647 Angelica gigas samples (282 imported and 365 domestic samples) was kindly supplied by the National Agricultural Products Quality Management Service (NAQS) in Seoul, Korea. These samples were collected over a period of 7 years. Both the sesame and Angelica gigas samples were ground into powders (20-mesh) for the collection of diffuse reflectance spectra. A NIR dataset composed of 48 diesel and 47 light gas oil (LGO) samples prepared in a previous publication [21] was also used in this study. LGO is produced by the fractional distillation of crude oil between 250 and 350 °C at atmospheric pressure and it is composed of hydrocarbons with approximately 10–21 carbon atoms per molecule. Transmission spectra were collected for the diesel and LGO samples with a pathlength of 10 mm. All of the NIR spectra were collected with a Foss NIRSystems Model 6500 spectrometer equipped with a quartz halogen lamp and PbS detector.

For the sesame dataset, 346 imported and 552 domestic samples were allocated to the calibration set. The validation set consisted of the remaining 86 imported and 138 domestic samples. The calibration set for the Angelica gigas was composed of 519 samples (226 imported and 293 domestic samples) and the rest of the samples were assigned to the validation set (56 imported and 72 domestic samples). Thirty-three diesel and 32 LGO samples were selected for the calibration set and the remaining samples were used for the validation set (15 diesel and 15 LGO samples). In all three cases, the calibration samples were randomly selected. All of the spectral pre-treatments as well as the discrimination analyses, including multiplicative scatter correction (MSC), baseline correction, PCA–LDA, PLS–DA, *k*-NN and HQI-voting, were done in Matlab R2011a (Math Works Inc., USA) using Matlab library functions such as pca, knnsearch, pls (pls toolbox). As an example, CPU time for assigning a sample to a certain class in the case of 128 Angelica gigas validation samples required 0.09–0.11 s when using a PC (AMD FX 4100 3.62 GHz, 3.0 RAM) equipped with a single core.

3. Results and discussion

3.1. Comparison of NIR spectral features between two groups in each dataset

Fig. 2(a) shows the average NIR spectra of the domestic (blue) and imported (red) sesame samples. Before averaging of the spectra in each group, all of the raw spectra were preprocessed using multiplicative scatter correction (MSC). As shown, the spectral features of the two groups were quite similar to each other and only minor spectral differences were observed in the 1800–2050 and 2300–2450 nm ranges. Fig. 2(b) presents the average NIR spectra of the Angelica gigas samples from domestic (blue) and imported (red) origins. In addition, the raw spectra were MSC-treated before averaging. In comparison with the average spectra of the sesame samples, the spectral difference between the Angelica gigas samples of two different origins was relatively more distinct. The elucidation of the botanical justification supporting the minute spectral differences in Fig. 2(a) and (b) would require highly extensive phytochemical investigation and, therefore, it is beyond the scope of this study. However, it is reasonable to believe that the difference in their spectral features is from the compositional variations caused by diverse factors such as different genetic origins and metabolic pathways. Based on the observation of the average spectra of both the sesame and Angelica gigas samples, the discrimination according to their geographical origin would not be an easy task due to the lack of significant spectral distinction. In this situation, a method able to effectively reflect minor spectral differences in discrimination analysis becomes more valuable.

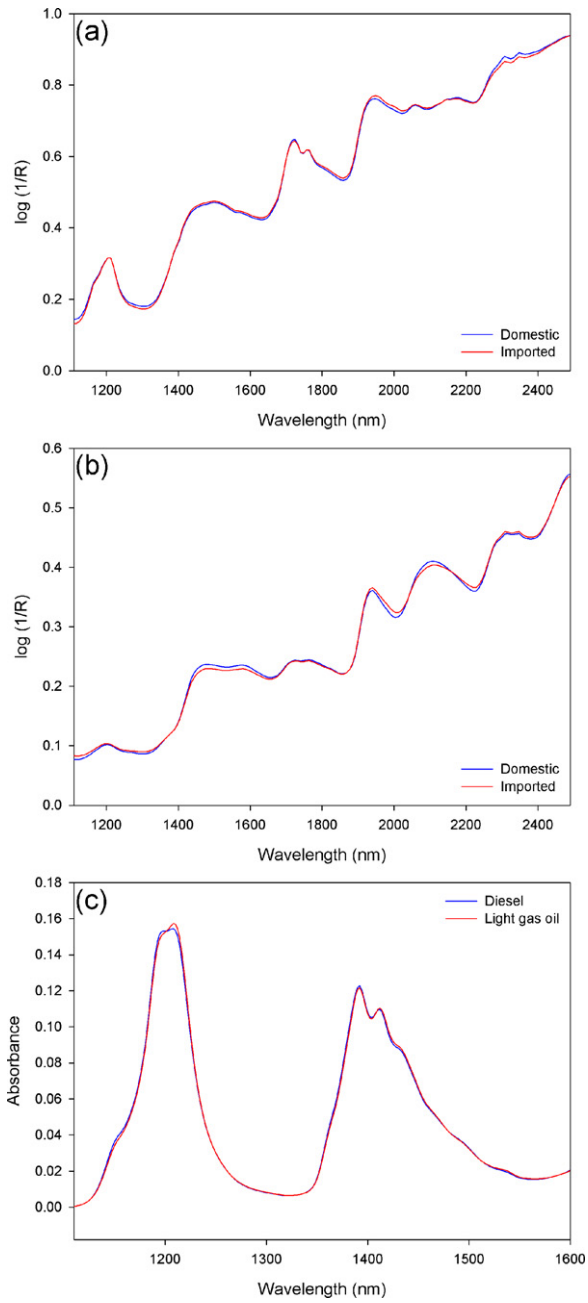

Fig. 2. Average diffuse reflectance NIR spectra of domestic (blue) and imported (red) sesame samples (a), and the average diffuse reflectance NIR spectra of Angelica gigas samples from domestic (blue) and imported (red) origins (b). The average transmission NIR spectra of diesel (blue) and LGO (red) are also shown (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

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Table 1
The discrimination errors acquired using PCA–LDA, PLS–DA, *k*-NN (euclidian), *k*-NN (standardized euclidian), *k*-NN (HQI) and HQI-voting for sesame, Angelica gigas, and diesel/LGO samples. The numbers in parentheses in the columns for the PCA–LDA and PLS–DA correspond to selected optimal two-factor combination and number of factors, respectively. The numbers in parentheses in the columns for *k*-NN (euclidian), *k*-NN (standardized euclidian), *k*-NN (HQI) and HQI-voting indicate the number of votes (*k*).

	PCA–LDA (%)	PLS–DA (%)	*k*-NN (euclidian) (%)	*k*-NN (standardized euclidian) (%)	*k*-NN(HQI) (%)	HQI-voting (%)
Sesame	13.4 (1st, 4th)	8.9 (3)	7.6 (7)	7.6 (7)	7.6 (7)	6.3 (8)
Angelica gigas	10.2 (2nd, 4th)	11.7 (2)	8.6 (1)	4.7 (1)	8.6 (1)	8.6 (1)
Diesel/LGO	6.7 (1st, 2nd)	6.7 (7)	3.3 (1)	3.3 (1)	0.0 (1)	0.0 (1)

Fig. 2(c) shows the average transmission NIR spectra of the diesel (blue) and LGO (red) samples in the 1100–1600 nm range. The overall spectral features of diesel and LGO are analogous since diesel is a product blended with 4–5 components, and LGO is the major constituent (approximately 80%) of diesel. However, fairly distinct spectral differences were observed at the 1194 and 1210 nm bands, corresponding to the second overtone of the CH_3 and CH_2 vibrations [22,23], respectively. The higher absorption at 1210 nm in the LGO spectrum indicates that it has less paraffin components than the diesel.

3.2. Discrimination analysis

PCA–LDA, PLS–DA, and HQI-voting were simultaneously used for discrimination of the samples listed in Fig. 1. For PCA–LDA, PCA was initially performed using each spectral dataset and the resulting scores were used for the LDA. A combination of two scores was employed since it was easy to visualize the discrimination performance in the two-dimensional domain. For the PLS–DA, discrimination models were developed by assigning one group as 1 and the other group as 2. The prediction accuracy was obtained by evaluating the predicted values of the samples, whether these were below or above 1.5. For *k*-NN, two different similarity measure functions of euclidian and standardized euclidian were evaluated and optimal *k* was determined at each case via cross-validation. In addition, *k*-NN was also performed using HQI as a similarity measure for comparison. HQI-voting was performed as described earlier in the introduction.

The discrimination errors obtained by predicting the samples in each validation set are summarized in Table 1. The discrimination error corresponds to the percentage of number of mis-predicted samples over total number of samples in the validation set. For the more detail comparison of discrimination performances of each method, the corresponding accuracy, sensitivity and specificity are summarized in supplementary data. The numbers in parentheses in the columns for the PCA–LDA and PLS–DA correspond to selected optimal two-factor combination and number of factors, respectively. The variation of discrimination error according to the combination of two factors in PCA–LDA and number of factors in PLS–DA is also presented in supplementary data. The numbers in parentheses in the columns for *k*-NN (euclidian), *k*-NN (standardized euclidian), *k*-NN (standardized euclidian) and HQI-voting indicate the number of votes (*k*). As shown in Table 1, the discrimination accuracies when HQI-voting was used improved in comparison with those resulted from PCA–LDA and PLS–DA for all three samples, while slightly better or equal to those obtained using *k*-NN. This result demonstrates that HQI-voting is a comparable and effective discrimination method as a variant of *k*-NN. Therefore, the discrimination results obtained with the use of PCA–LDA, PLS–DA and HQI-voting are mainly analyzed further in detail.

Fig. 3(a) shows the score scatter plot (the first vs. fourth scores) that resulted from the use of PCA–LDA for the discrimination between domestic (blue circle) and imported (red circle) sesame samples. The boundary (dotted line) between these two groups that was determined by LDA is also displayed. The PLS–DA prediction plot for the same discrimination data is presented in Fig. 3(b). The samples located in the bottom-left and top-right boxes were correctly predicted, while those in the top-left and bottom-right boxes were incorrectly predicted.

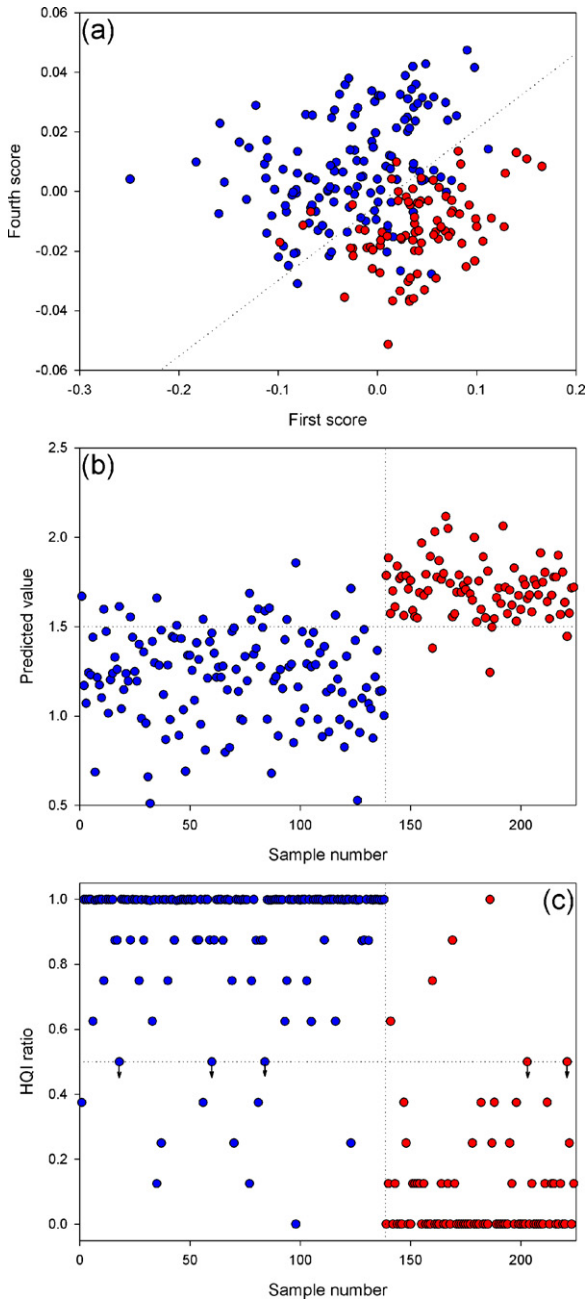

Fig. 3. Score scatter plot (the first vs. fourth scores) resulting from the use of PCA–LDA (a), the PLS–DA prediction plot (b), and the HQI-ratio plot (c) for the discrimination between the domestic (blue circles) and imported (red circles) sesame samples. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

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The HQI-ratio was used in order to graphically present the discrimination results using HQI-voting. This is the ratio between the sum of the HQI values from all of the votes (denominator) and the sum of the HQI values that voted for domestic samples only (numerator). A HQI ratio of 0 (zero) indicated that there were no votes for a domestic sample, which indicated a unanimous decision for an imported sample. On the contrary, a HQI ratio of 1 indicated a unanimous decision for a domestic sample. Split decisions resulted in HQI ratios other than 0 or 1, and a HQI ratio below or above 0.5 indicated the decision was for an imported or domestic sample, respectively. The HQI ratio plot for the discrimination of sesame samples is shown in Fig. 3(c). The samples positioned in the bottom-left or top-right boxes were incorrectly predicted as an imported or domestic group, respectively. The resulting discrimination accuracy was 6.3% as seen in Table 1. However, it degraded to 8.3% when only the first vote was evaluated. As previously discussed, the spectral difference between the domestic and imported sesame samples is minute, so simultaneous evaluation of the decisions from multiple votes could be more robust.

The score scatter, PLS–DA prediction, and HQI ratio plots for the discrimination of the Angelica gigas and diesel/LGO samples are shown in Figs. 4 and 5, respectively. The blue and red circles in Fig. 4 indicate the domestic and imported Angelica gigas samples, respectively. The blue and red circles in Fig. 5 correspond to the diesel and LGO samples, respectively. In both of the HQI ratio plots, the results are either values of 1 or 0, because only one vote corresponding to the highest HQI value was used for the corresponding discriminations. The spectral differences between the two groups of Angelica gigas samples as well as the LGO/diesel samples are relatively apparent as shown in Fig. 2(b) and (c). Therefore, the adoption of only the first (highest ranked) vote for the determination could be sufficient.

The samples resulting in non-unanimous decisions among 8 votes were selected to investigate the difference in decisions made by evaluating either a single vote or multiple votes for the discrimination of sesame samples. Out of a total of 224 validation samples, 72 samples showed split decisions (40 domestic and 32 imported samples). Fig. 6 shows the results of non-unanimous 8 individual votes made for the determination of sesame sample group. Each decision is indicated by either blue (domestic) or red (imported) colors. The boxes (a) and (b) show the results obtained for domestic and imported sesame samples, respectively. For example, the blue color in box (a) implies correct prediction of the group and the same color in box (b) corresponds to incorrect prediction of the group.

The D_8 and D_1 columns indicate the group decision by simultaneous evaluation of 8 votes and evaluation of the first vote only, respectively. If the colors in the D_8 and D_1 columns are the same, then there is no reversal of decision between the two voting methods. By contrast, different colors in these columns indicate contradictory decisions from the two different voting methods. The decisions were maintained for most of the samples. However, 9 samples showed a reversal of decision by evaluating the 8 votes indicated by arrows in Fig. 6. Among them, 7 samples were mis-predicted by evaluation of only the first vote, but were then correctly predicted through the 8-vote evaluation (indicated by circles). In addition, 2 of the samples predicted correctly by evaluation of the first vote were mis-predicted by the 8-vote evaluation (indicted by crosses). In total, additional 5 samples were predicted correctly through the 8-vote evaluation. The simultaneous evaluation of multiple votes was helpful in order to improve the discrimination accuracy especially when the spectral difference between the two groups was minute, such as the NIR spectra of the sesame samples in this study.

In order to investigate the prediction characteristics of the PCA–LDA, PLS–DA, and HQI-voting further, the incorrectly predicted samples for the discrimination of sesame samples were compared for each method. Fig. 7 shows the box plots that show all of the mis-predicted samples when PCA–LDA, PLS–DA, and HQI-voting were used for the discrimination. The dark squares indicate the mis-predicted samples in each case and the numbers on top of each column designate the sample number in the validation set. The results indicate that HQI-voting primarily improves the discrimination accuracy via correctly predicting the groups of samples that were incorrectly predicted when either PCA–LDA or PLS–DA was used. It implies that the recognition of spectral features for discrimination using HQI-voting is comparable to the existing factor-based discrimination methods.

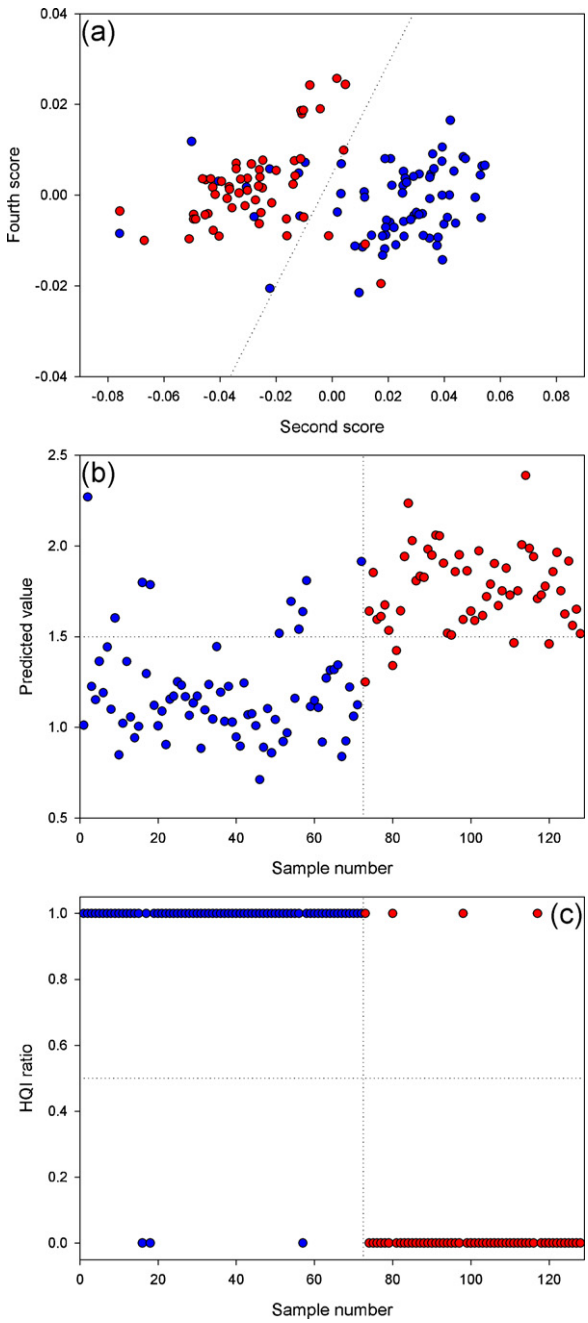

Fig. 4. Score scatter plot (the second vs. fourth scores) resulting from the use of PCA–LDA (a), the PLS–DA prediction plot (b), and the HQI-ratio plot (c) for the discrimination between the domestic (blue circles) and imported (red circles) Angelica gigas samples. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

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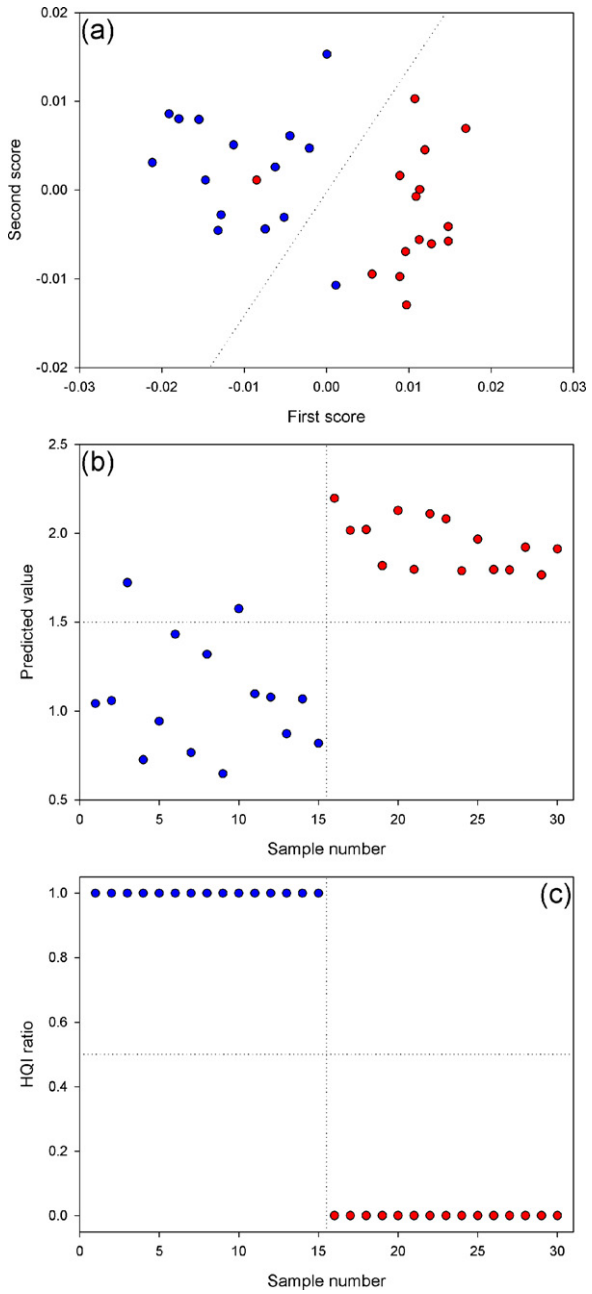

Fig. 5. Score scatter plot (the first vs. second scores) resulting from the use of PCA–LDA (a), the PLS–DA prediction plot (b), and the HQI-ratio plot (c) for the discrimination between the diesel (blue circles) and LGO (red circles) samples. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

3.3. Variation of discrimination accuracy according to the number of calibration samples

In order to explore the characteristics of HQI-voting further, the variation of discrimination accuracy was examined by systematically changing the number of calibration samples. The sesame dataset was used again for this evaluation. The number of calibration samples varied from 10 to 898 (increment interval: 74) and the corresponding discrimination errors were obtained. The samples in the prediction set were unchanged. In each case, the samples were randomly selected out of all of the samples in the original calibration set and the random sample selection was repeated 200 times for cross-validation.

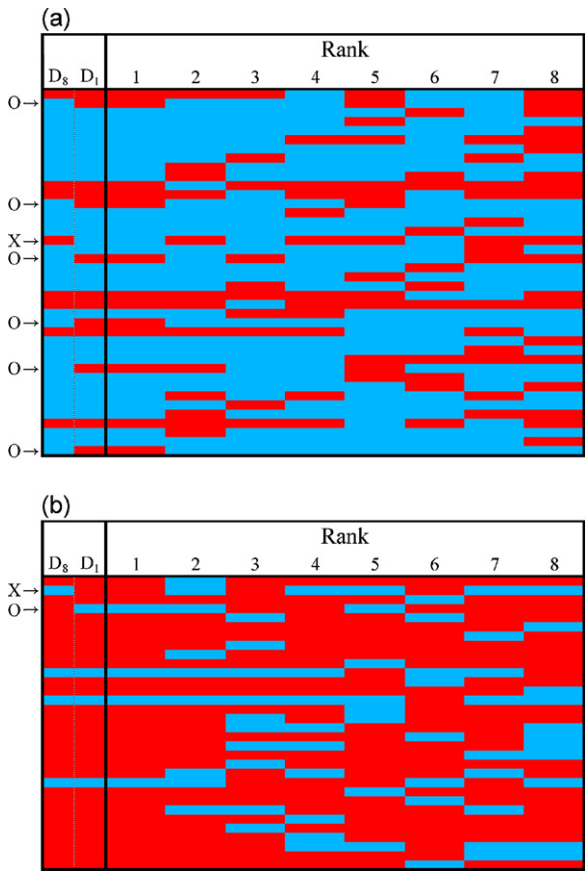

Fig. 6. Results of non-unanimous 8 individual votes made for the determination of sesame sample group. Boxes (a) and (b) show the prediction results for the domestic and imported sesame samples, respectively. Each decision is indicated by either blue (domestic) or red (imported) colors. D_8 and D_1 indicate the group decision by the simultaneous evaluation of 8 votes and evaluation of only the first vote, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Fig. 8 shows the variation in the errors for the discrimination of the sesame samples using PCA–LDA, PLS–DA and HQI-voting when the number of calibration samples varied. Although the number of calibration samples varied, the ratios between number of domestic and imported sesame samples maintained constant for each case (40% imported and 60% domestic samples in each dataset). The error in each case was the result of averaging 200 cross-validated errors and the error bar indicates the corresponding standard deviation. The trends in the variation of the discrimination errors were quite analogous for the three methods. The errors decreased substantially until the number of samples reached 84 or 158 and then the improvement in accuracy was rather minor or insignificant after that point.

When PCA–LDA and PLS–DA were used, the improvement in the accuracies was nearly insignificant after a certain point even with the continual addition of calibration samples. In contrast, the discrimination error gradually decreased with the addition of calibration samples in the case of HQI-voting. Since the sample selection was random, the selected samples could possibly represent wide compositional variation even though the number of samples incorporated for calibration was not large. Therefore, the inclusion of additional samples for calibration beyond a certain stage would not help to improve the discrimination accuracy in the case of using either PCA–LDA or PLS–DA as shown in Fig. 8. In contrast, HQI-voting is based on direct spectral matching, so the increased sample population could be advantageous in order to determine the better match for an unknown sample. This fact

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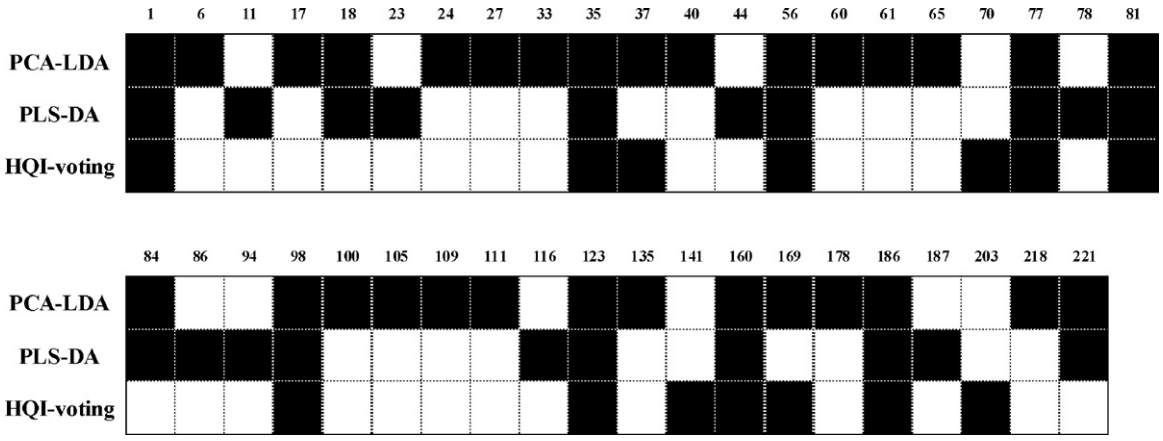

Fig. 7. Box plots showing all of the mis-predicted samples when PCA–LDA, PLS–DA, and HQI-voting are used for the discrimination of sesame samples. The dark squares indicate the mis-predicted samples in each case and the numbers on top of each column indicate the sample number in the validation set.

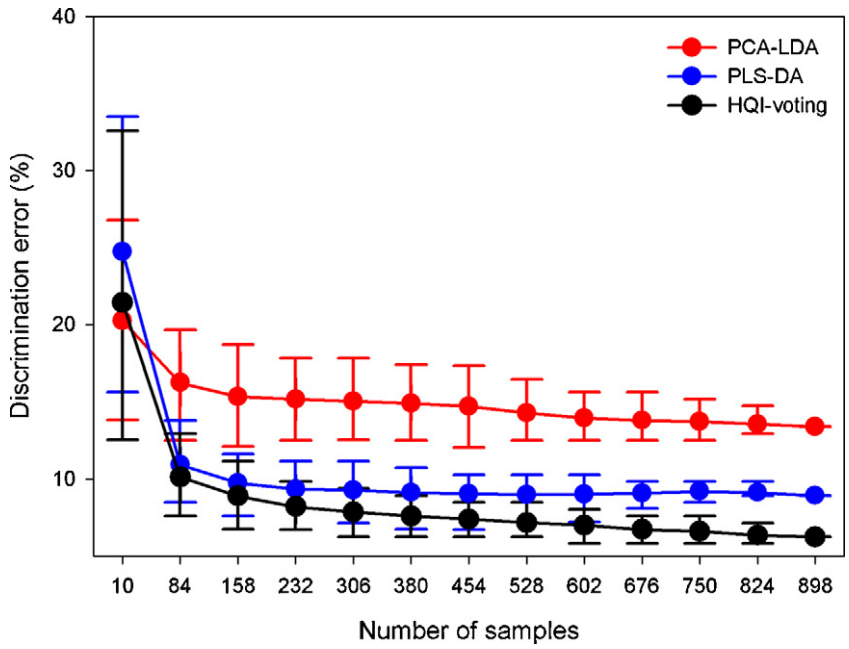

Fig. 8. Variation of errors for the discrimination of sesame samples using PCA–LDA, PLS–DA, and HQI-voting when the number of calibration samples varies. The error in each case is a result of averaging 200 cross-validated errors and the error bar indicates the corresponding standard deviation.

explains the continual improvement of the discrimination accuracy in HQI-voting with an increase in the population of calibration samples.

4. Conclusion

The utility of HQI-voting has been demonstrated for NIR spectroscopic discrimination of agricultural samples according to their geographical origins and two similar petroleum products. The fact that no modeling steps are needed without the worry of overfitting is the most practical advantage of HQI-voting when it is adopted in the field, since many field analysts prefer simple methods able to transparently understand and easily apply for routine analysis without taking many complex variables into consideration. The improved discrimination performance when using HQI-voting cannot be generalized since only three case studies have been presented here. However, we think that it has strong potential as an alternative discrimination method in parallel with existing factor-based multivariate methods. In the future, HQI-voting will be further evaluated for discrimination of samples using other spectroscopic methods. In addition, the development of a strategy utilizing HQI for quantitative analysis is under way.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.aca.2012.10.058.

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